

Digital Pendulum - Installation & Commissioning

33-936-0V65

Feedback



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Technology Training for tomorrow's world

Control in a MATLAB® Environment

Digital Pendulum

Installation and Commissioning

33-936-0V65

**(For use with MATLAB 6.5 version
and Windows 95, 98, NT, 2000)**



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Notes



DIGITAL PENDULUM SYSTEM INSTALLATION AND COMMISSIONING

Preface

THE HEALTH AND SAFETY AT WORK ACT 1974

We are required under the Health and Safety at Work Act 1974, to make available to users of this equipment certain information regarding its safe use.

The equipment, when used in normal or prescribed applications within the parameters set for its mechanical and electrical performance, should not cause any danger or hazard to health or safety if normal engineering practices are observed and they are used in accordance with the instructions supplied.

If, in specific cases, circumstances exist in which a potential hazard may be brought about by careless or improper use, these will be pointed out and the necessary precautions emphasised.

While we provide the fullest possible user information relating to the proper use of this equipment, if there is any doubt whatsoever about any aspect, the user should contact the Product Safety Officer at Feedback Instruments Limited, Crowborough.

This equipment should not be used by inexperienced users unless they are under supervision.

We are required by European Directives to indicate on our equipment panels certain areas and warnings that require attention by the user. These have been indicated in the specified way by yellow labels with black printing, the meaning of any labels that may be fixed to the instrument are shown below:



CAUTION -
RISK OF
DANGER

Refer to accompanying documents



CAUTION -
RISK OF
ELECTRIC SHOCK



CAUTION -
ELECTROSTATIC
SENSITIVE DEVICE

PRODUCT IMPROVEMENTS

We maintain a policy of continuous product improvement by incorporating the latest developments and components into our equipment, even up to the time of dispatch.

All major changes are incorporated into up-dated editions of our manuals and this manual was believed to be correct at the time of printing. However, some product changes which do not affect the instructional capability of the equipment, may not be included until it is necessary to incorporate other significant changes.

COMPONENT REPLACEMENT

Where components are of a 'Safety Critical' nature, i.e. all components involved with the supply or carrying of voltages at supply potential or higher, these must be replaced with components of equal international safety approval in order to maintain full equipment safety.

In order to maintain compliance with international directives, all replacement components should be identical to those originally supplied.

Any component may be ordered direct from Feedback or its agents by quoting the following information:

- | | |
|------------------------|----------------------------|
| 1. Equipment type | 2. Component value |
| 3. Component reference | 4. Equipment serial number |

Components can often be replaced by alternatives available locally, however we cannot therefore guarantee continued performance either to published specification or compliance with international standards.



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DECLARATION CONCERNING ELECTROMAGNETIC COMPATIBILITY

Should this equipment be used outside the classroom, laboratory study area or similar such place for which it is designed and sold then Feedback Instruments Ltd hereby states that conformity with the protection requirements of the European Community Electromagnetic Compatibility Directive (89/336/EEC) may be invalidated and could lead to prosecution.

This equipment, when operated in accordance with the supplied documentation, does not cause electromagnetic disturbance outside its immediate electromagnetic environment.

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DIGITAL PENDULUM SYSTEM INSTALLATION AND COMMISSIONING

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1 Introduction

This manual contains a description of the components of the Digital Pendulum System, assembly instructions and test procedures. It is intended to provide the means for validating the correct operation of the system, **before** proceeding to more advanced functions and demonstrations using control software with MATLAB / SIMULINK

The procedures described in this section of the Pendulum manual will ensure correct:

- Assembly of the system
- Setting up of the EISA or PCI Advantech Board
- Cable connections between the PC and the Pendulum

1.1 Required Equipment

- | | | | | | | | |
|-------------|--|-------------|--|-------------|---|-------------|--------------------------------------|
| 1 | Software and PC system as described in the manual 33-000-0C - Introduction and Computer Systems Installation | | | | | | |
| 2 | Set of 3 "Control in a MATLAB Environment" Manuals: <table border="0" style="margin-left: 40px;"><tr><td>33-000-0V65</td><td>Introduction and Software Installation</td></tr><tr><td>33-936-0V65</td><td>Digital Pendulum - Installation and Commissioning</td></tr><tr><td>33-936-1V65</td><td>Digital Pendulum Control Experiments</td></tr></table> | 33-000-0V65 | Introduction and Software Installation | 33-936-0V65 | Digital Pendulum - Installation and Commissioning | 33-936-1V65 | Digital Pendulum Control Experiments |
| 33-000-0V65 | Introduction and Software Installation | | | | | | |
| 33-936-0V65 | Digital Pendulum - Installation and Commissioning | | | | | | |
| 33-936-1V65 | Digital Pendulum Control Experiments | | | | | | |
| 3 | Feedback Supplied equipment as follows <table border="0" style="margin-left: 40px;"><tr><td>33-200</td><td>Digital Pendulum Mechanical Unit</td></tr><tr><td>33-201</td><td>Digital Pendulum Controller</td></tr></table> | 33-200 | Digital Pendulum Mechanical Unit | 33-201 | Digital Pendulum Controller | | |
| 33-200 | Digital Pendulum Mechanical Unit | | | | | | |
| 33-201 | Digital Pendulum Controller | | | | | | |



DIGITAL PENDULUM SYSTEM INSTALLATION AND COMMISSIONING

Chapter 1 Introduction

4	33-936	Digital Pendulum Control System Software pack including Advantech PCI1711 card
	33-936I	Digital Pendulum Control System Software pack excluding Advantech PCI1711 card

Or the composite products (including items 3 and 4 above)

33-005-PCI	Contains 33-200 , 33-201 , 33-936 (Composite product)
33-005I-PCI	Contains 33-200 , 33-201 , 33-936I (Composite product)



2 Mechanical Assembly

Carefully unpack the items, and remove all packaging and securing material. You should have the following items:

- 1) Track for the cart to move on, with motor attached, and toothed belt
- 2) Pendulum cart with mounting points for the two pendulum arms and attached ribbon cable
- 3) Two pendulum arms with weights attached
- 4) Two legs each with two adjustable feet, with mounting plates for securing to the track
- 5) 1 x 2 mm and 1 x 6 mm Allen keys

Refer to Figure 2.1 below

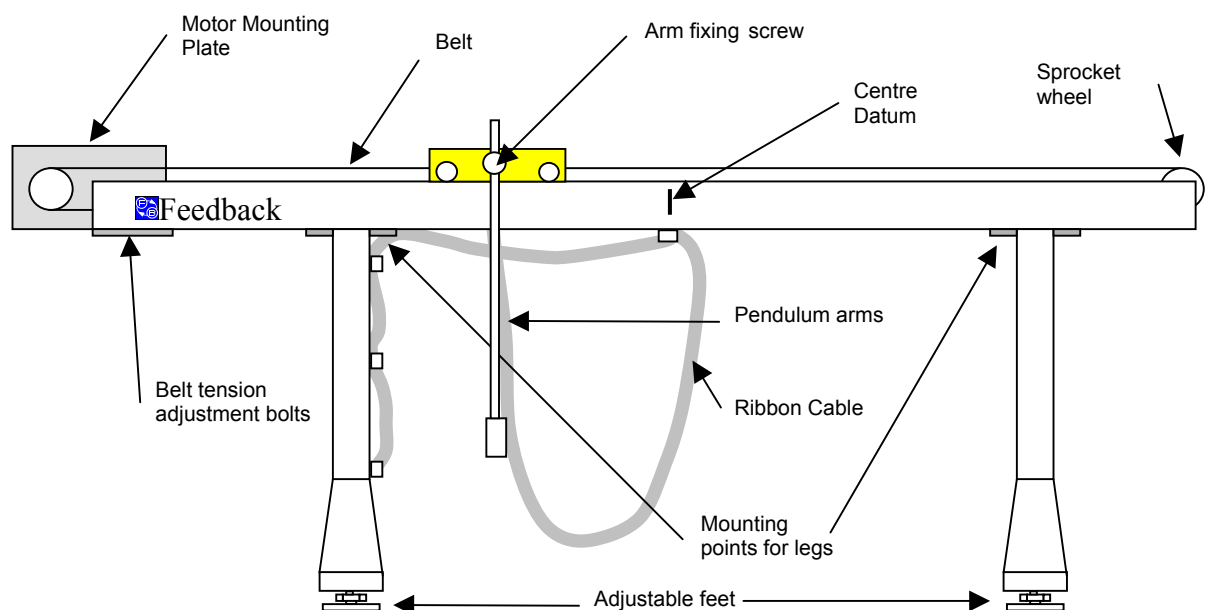


Fig 2.1: The pendulum system

Place the Digital Pendulum Mechanical Unit on a level surface with the motor on the left viewed from the front and allow sufficient room for the controller box, PC and mouse. Ensure that the pendulum is clear of any obstructions when executing a complete revolution, and is not able to make contact with any observers.

Using the 6mm Allen key provided, remove the 8 bolts attached to the track at the leg mounting positions, and secure the legs to the track with these bolts.



Use the adjustable feet (it is only necessary to adjust one of the four) to ensure that the pendulum system is stable on the surface. Slacken off the adjustable foot locking nut and rotate the foot until the system is perfectly stable. Tighten the locking nut.

Insert one of the pendulum arms into one of the two mounting positions on the cart from the bottom and loosely tighten the securing screw on to the flat part of the pendulum arm with the 2 mm Allen key provided. Push the arm upwards until the end of the flat is reached and tighten the securing screw. Repeat for the other pendulum arm on the other side of the cart

Ensure the pendulum cart is free to move on the track and that the drive belt is attached to the cart at both ends and goes over the sprocket wheels at each end of the track. If the belt is slack, use the 6 mm Allen key to slacken off the belt tension adjustment bolts and move the motor mounting plate until the belt is taut. Tighten the adjustment bolts.

Finally, clip the ribbon cable to the pendulum left hand leg using the attached clips.



3 Connections to the Computer

Following the installation of the interface card into your PC, you are now ready to connect the ribbon cables to the 33-201 Digital Pendulum Controller.

Refer to Figures 3.1 and 3.2 below for details.

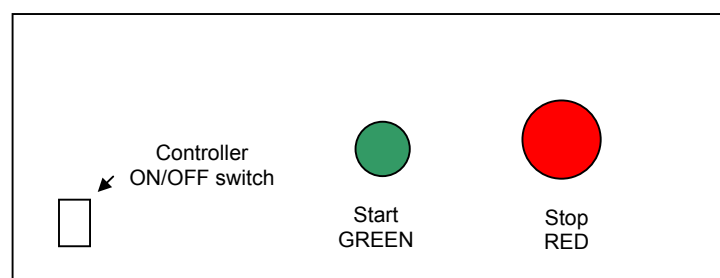


Figure 3-1: Front view of 33-201 controller

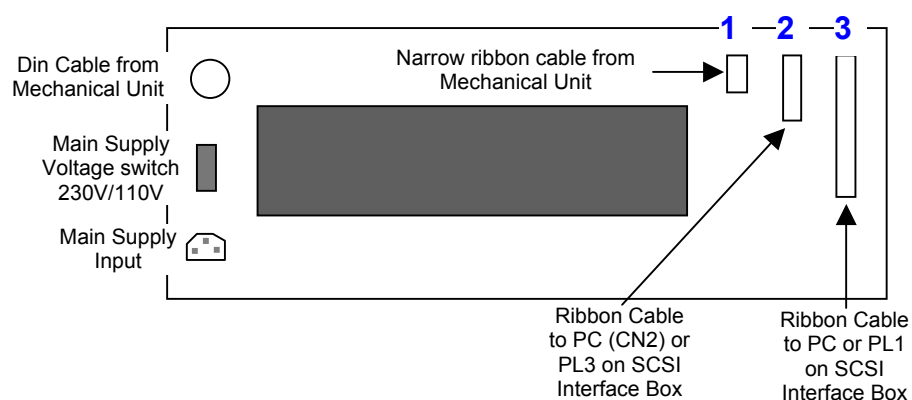


Figure 3-2: Rear view of 33-201 controller



PCI1711 PCI

Connect the **very** narrow ribbon cable from the Pendulum Mechanical Unit to connector 1 in Figure 3.4.

Connect the round SCSI cable from the PCI1711 board to the SCSI connector on the SCSI Adapter Box.

Connect the narrow ribbon cable from connector **PL3** on the SCSI Adapter Box to connector **2** in Figure 3.4. Connect the wide ribbon cable from connector **PL1** on the SCSI Adapter box to connector **3** in Figure 3.4.

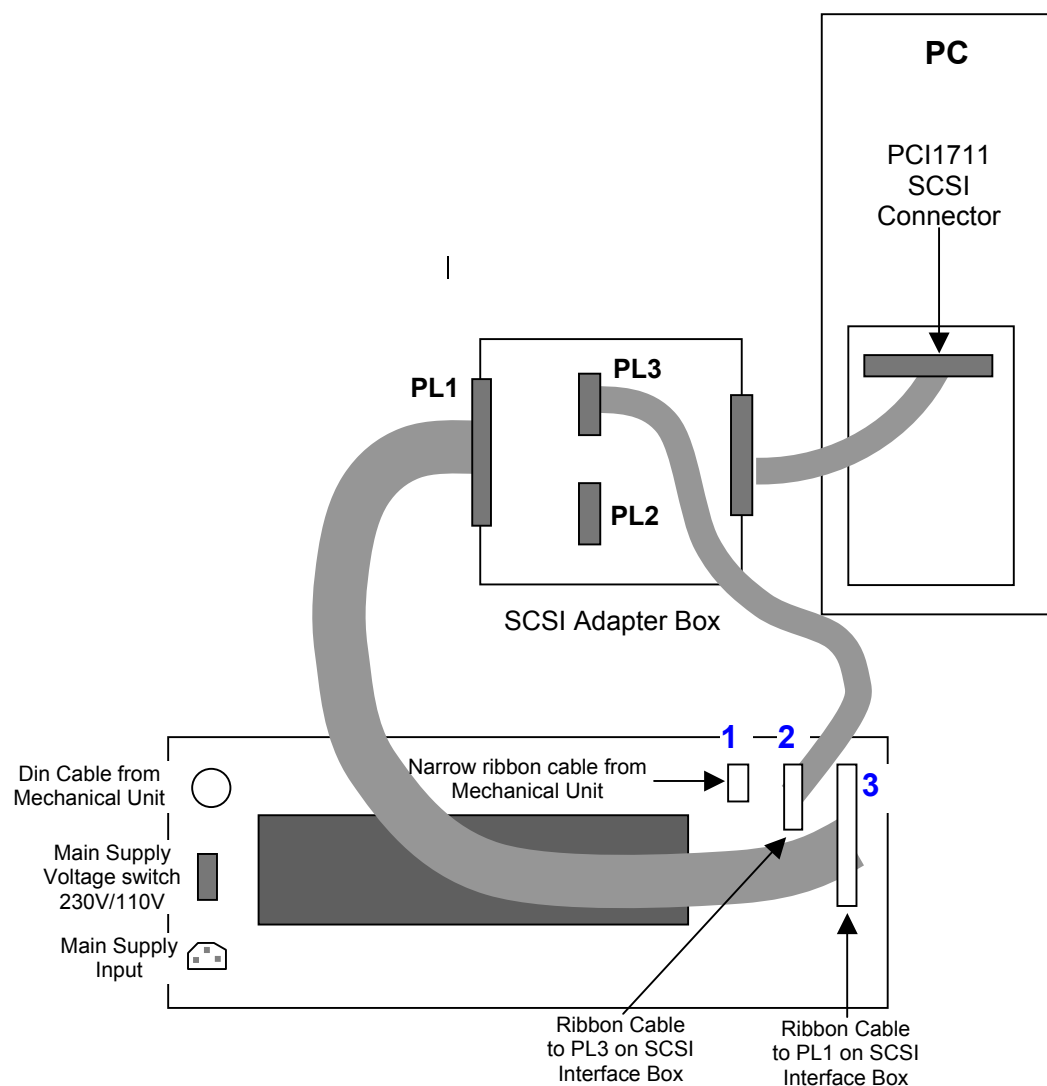


Figure 3-3: Cable connections to Advantech PCI1711 Board



DIGITAL PENDULUM SYSTEM INSTALLATION AND COMMISSIONING

Chapter 3

Connections to the Computer

- Voltage selector** Set the voltage selector for the correct voltage in your area.
- Mains power** Connect the mains power cable to the 33-201 and plug in to your mains supply.
- Switch on** Switch on your PC and the mains supply to the 33-201.
- Switch the 33-201 illuminated power switch on. If this switch does not light up check the mains power cable and mains power supply. Do not at this stage press the round green start button.



It is important that the area of operation is free of obstruction and that users and observers are warned to keep clear of the immediate vicinity of the system for safety reasons.



**DIGITAL PENDULUM SYSTEM
INSTALLATION AND COMMISSIONING**

Chapter 3

Connections to the Computer

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